

Point	x-coordinate	y-coordinate
A	$-\frac{4\pi}{3} \approx -4.2$	$-\frac{2\pi}{3} + \frac{\sqrt{3}}{2} \approx -1.2$
B	$-\frac{2\pi}{3} \approx -2.1$	$-\frac{\pi}{3} + \left(-\frac{\sqrt{3}}{2}\right) \approx -1.9$
C	$\frac{2\pi}{3} \approx 2.1$	$\frac{\pi}{3} + \frac{\sqrt{3}}{2} \approx 1.9$
D	$\frac{4\pi}{3} \approx 4.2$	$\frac{2\pi}{3} + \left(-\frac{\sqrt{3}}{2}\right) \approx 1.2$

EXERCISES 4.3

Exer. 1–16: Find the local extrema of f and the intervals on which f is increasing or is decreasing, and sketch the graph of f .

- 1 $f(x) = 5 - 7x - 4x^2$
- 2 $f(x) = 6x^2 - 9x + 5$
- 3 $f(x) = 2x^3 + x^2 - 20x + 1$
- 4 $f(x) = x^3 - x^2 - 40x + 8$
- 5 $f(x) = x^4 - 8x^2 + 1$
- 6 $f(x) = 4x^3 - 3x^4$
- 7 $f(x) = 10x^3(x - 1)^2$
- 8 $f(x) = (x^2 - 10x)^4$
- 9 $f(x) = x^{4/3} + 4x^{1/3}$
- 10 $f(x) = x(x - 5)^{1/3}$
- 11 $f(x) = x^{2/3}(x - 7)^2 + 2$
- 12 $f(x) = x^{2/3}(8 - x)$
- 13 $f(x) = x^2\sqrt{x^2 - 4}$
- 14 $f(x) = 8 - \sqrt[3]{x^2 - 2x + 1}$
- 15 $f(x) = x\sqrt{x^2 - 9}$
- 16 $f(x) = x\sqrt{4 - x^2}$

Exer. 17–22: Find the local extrema of f on $[0, 2\pi]$ and the subintervals on which f is increasing or is decreasing. Sketch the graph of f .

- 17 $f(x) = \cos x + \sin x$
- 18 $f(x) = \cos x - \sin x$
- 19 $f(x) = \frac{1}{2}x - \sin x$
- 20 $f(x) = x + 2 \cos x$
- 21 $f(x) = 2 \cos x + \sin 2x$
- 22 $f(x) = 2 \cos x + \cos 2x$

Exer. 23–28: Find the local extrema of f .

- 23 $f(x) = \sqrt[3]{x^3 - 9x}$
- 24 $f(x) = \sqrt{x^2 + 4}$
- 25 $f(x) = (x - 2)^3(x + 1)^4$
- 26 $f(x) = x^2(x - 5)^4$
- 27 $f(x) = \frac{\sqrt{x - 3}}{x^2}$
- 28 $f(x) = \frac{x^2}{\sqrt{x + 7}}$

Exer. 29–32: Find the local extrema of f on the given interval.

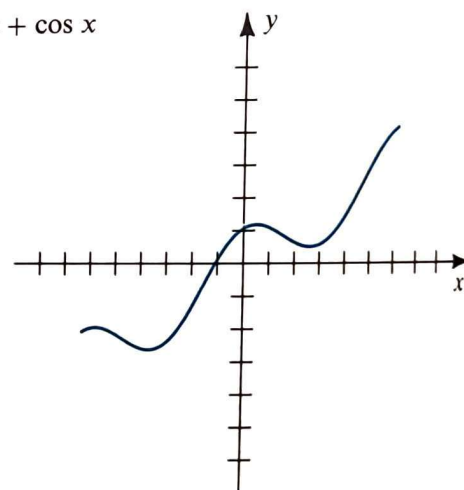
- 29 $f(x) = \sec \frac{1}{2}x$; $[-\pi/2, \pi/2]$
- 30 $f(x) = \cot^2 x + 2 \cot x$; $[\pi/6, 5\pi/6]$

31 $f(x) = 2 \tan x - \tan^2 x$; $[-\pi/3, \pi/3]$

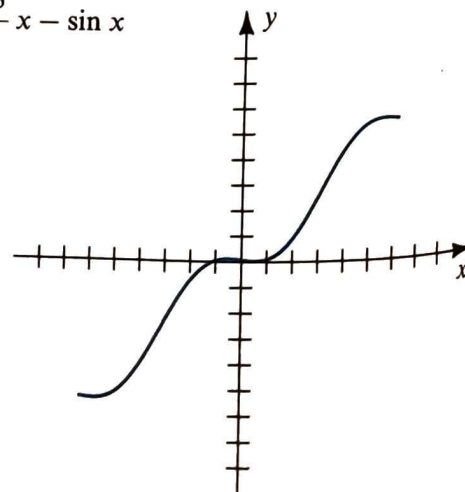
32 $f(x) = \tan x - 2 \sec x$; $[-\pi/4, \pi/4]$

Exer. 33–34: Shown in the figure is a graph of the equation for $-2\pi \leq x \leq 2\pi$. Determine the x -coordinates of the points that correspond to local extrema.

33 $y = \frac{1}{2}x + \cos x$



34 $y = \frac{\sqrt{3}}{2}x - \sin x$



Exer. 35–40: Sketch the graph of a differentiable function f that satisfies the given conditions.

35 $f(0) = 3$; $f(-2) = f(2) = -4$; $f'(0)$ is undefined;
 $f'(-2) = f'(2) = 0$; $f'(x) > 0$ if $-2 < x < 0$ or $x > 2$;
 $f'(x) < 0$ if $x < -2$ or $0 < x < 2$

36 $f(3) = 5$; $f(5) = 0$; $f'(5)$ is undefined; $f'(3) = 0$;
 $f'(x) > 0$ if $x < 3$ or $x > 5$; $f'(x) < 0$ if $3 < x < 5$

37 $f(3) = 5$; $f(5) = 0$; $f'(3) = f'(5) = 0$;
 $f'(x) > 0$ if $x < 3$ or $x > 5$; $f'(x) < 0$ if $3 < x < 5$

38 $f(0) = 3$; $f(-2) = f(2) = -4$;
 $f'(-2) = f'(0) = f'(2) = 0$;
 $f'(x) > 0$ if $-2 < x < 0$ or $x > 2$;
 $f'(x) < 0$ if $x < -2$ or $0 < x < 2$

39 $f(-5) = 4$; $f(0) = 0$; $f(5) = -4$;
 $f'(-5) = f'(0) = f'(5) = 0$;
 $f'(x) > 0$ if $|x| > 5$; $f'(x) < 0$ if $0 < |x| < 5$

40 $f(a) = a$ and $f'(a) = 0$ for $a = 0, \pm 1, \pm 2, \pm 3$;
 $f'(x) > 0$ for all other values of x

c Exer. 41–42: Graph f on $[-2, 2]$. (a) Use the graph to estimate the local extrema of f . (b) Estimate where f is increasing or is decreasing.

41 $f(x) = \frac{x^2 - 1.5x + 2.1}{0.3x^4 + 2.3x + 2.7}$ 42 $f(x) = \frac{10 \cos 2x}{x^2 + 4}$

c Exer. 43–44: Graph f' on $[-1, 1]$. Estimate the x -coordinates of the local extrema of f and classify each local extremum.

43 $f'(x) = x - \cos \pi x - \sin x$

44 $f'(x) = 6x^3 - 3x^2 - 1.3x + 0.5$

4.4 CONCAVITY AND THE SECOND DERIVATIVE TEST

In the preceding section we used the sign of the derivative f' to determine where a function f was increasing and where it was decreasing. Similarly, we can use the sign of the *second* derivative f'' to determine where the *derivative* f' is increasing and where it is decreasing. Specifically, if $f''(x) > 0$ on an open interval I , then $f'(x)$ is increasing on I ; that is, the slope of the tangent line to the graph of f increases as x increases (see Figure 4.27). As the next definition states, the graph is *concave upward* on I .

FIGURE 4.27 Concave upward graph

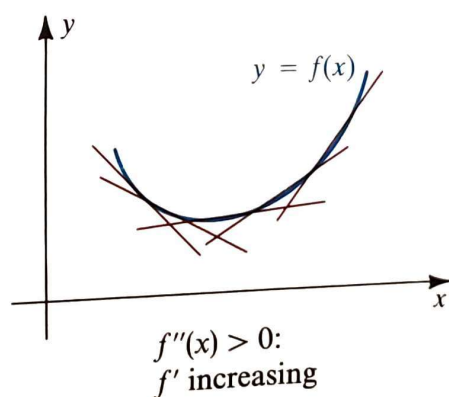
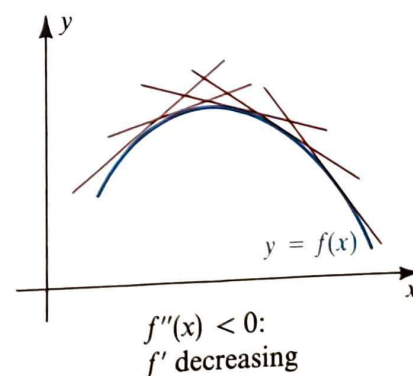


FIGURE 4.28 Concave downward graph



If $f''(x) < 0$ on an open interval I , then $f'(x)$ is decreasing on I . In this case, the slope of the tangent line to the graph of f decreases as x increases (see Figure 4.28), and we say that the graph is *concave downward* on I .

Definition (4.15)

- Let f be differentiable on an open interval I . The graph of f is
- (i) **concave upward** on I if f' is increasing on I
 - (ii) **concave downward** on I if f' is decreasing on I